

QS-Net / CQ-ZDR on UNSW-NB15: Results Analysis, Pros, Cons, and Caveats

1 Run summary

Table 1: Configuration and headline numbers, UNSW-NB15 run.

Classes kept (8)	Normal, Exploits, Fuzzers, Generic, Reconnaissance, DoS, Analysis, Backdoor
Raw feature columns (8)	<code>sttl</code> , <code>dttl</code> , <code>dload</code> , <code>ct_state_ttl</code> , <code>state</code> , <code>dinpkt</code> , <code>ackdat</code> , <code>tcprtt</code>
N_{qubits} , layers, re-upload	8, 3, yes
Train / test / cal / zero-day n	3590 / 4000 / 4000 / 1500
Per-class cap (train)	500 (Analysis: 333, Backdoor: 257 – see §3)
Epochs	10

This run used an **earlier pipeline version**: Proposition 1 is checked with synthetic Ginibre-random density matrices (`check_proposition1`), not the data-only `check_proposition1_on_data` variant, and `F_max/attacks` are not yet routed through a `ForwardCache` or the gradient-once FGSM sweep. The numbers below are still valid, but are not yet produced under the same, fully-optimized code path used for the CICIoT run discussed previously – re-running under the latest notebook is listed as a recommendation in §??.

2 Pros

- **Proposition 1 (noise contraction) holds to machine precision**: max residual 4.44×10^{-16} across $p \in [0, 1]$ – an exact identity check, unaffected by which dataset is used.
- **Proposition 3 (conformal coverage) tracks the target α well**. The achieved false-alarm rate stays within the expected finite-sample band of the target for three of four α values:

target α	q	achieved FAR	n_{cal}
0.01	0.6708	0.0073	4000
0.05	0.4055	0.0510	4000
0.10	0.3448	0.0953	4000
0.20	0.2510	0.2168	4000

and the calibration/test exchangeability check (AUROC = 0.507) supports the distribution-free coverage guarantee’s assumption.

- **The certified machinery (Lipschitz bound, certified radius) behaves exactly as designed:** the analytic bound $L_\varphi \leq 1.5$ is never violated by the empirical tightness diagnostic (max sampled ratio 0.80–0.93 across all four splits), and the certified-radius computation runs end-to-end (mean $R = 0.064$ over 2135/4000 correctly- classified, accepted test samples).
- **Empirical adversarial degradation is graceful, not a cliff:** acceptance rate stays at 100% up to $\varepsilon_{L_2} \approx 0.14$, then decays smoothly to 63% by $\varepsilon_{L_2} = 0.85$ (breakpoint at 0.226). H3 (empirical breakpoint $\geq$ certified ε^*) nominally holds, subject to the vacuity caveat in §3.
- **The barren-plateau false-alarm pattern replicates:** the pre-training probe again flags risk (var = 9.5×10^{-9}), while actual training gradients stay healthy throughout (4.7×10^{-3} to 1.0×10^{-2}). Seeing this twice, on two different datasets, strengthens the conclusion that the probe is a weak heuristic here, not a real risk signal for this ansatz/init combination.
- **Algorithm 3’s mechanics execute correctly end-to-end** (Day-22 walk-through): fidelity $\rightarrow$ score $\rightarrow$ conformal test $\rightarrow$ label all compute as specified, even though the *decision quality* on this dataset is poor (§3).

3 Cons / concerns

- **Zero-day detection effectively fails on this dataset:** TPR = 0.012 (1.2%), versus FPR = 0.051 (on target). The detector accepts the held-out class as a known class *almost every time*. This traces directly to Proposition-2 quantities:

$$F_{\text{out}} = 0.979, \quad F_{\text{in}} = 0.202, \quad \Delta = -0.959, \quad \varepsilon^* = -0.323,$$

a far more severe failure than the companion CICIoT run ($F_{\text{out}} = 0.335$, $\varepsilon^* = -0.105$). Critically, the $\beta=0.05$ quantile-relaxed F_{out} is *identical* to the strict worst case (0.979 either way) – this is not one outlier zero-day sample dragging down an otherwise-good separation, it is the bulk of the zero-day distribution (mean $F_{\text{max}} = 0.912$, 95th percentile = 0.979) sitting almost on top of known-class prototypes. The Day-22 worked example shows this concretely: the zero-day sample gets $F_{\text{max}} = 0.881$ against *Fuzzers* and is accepted, not rejected.

- **Test accuracy (59%) hides a near-total collapse on three of eight classes:**

class	precision	recall	F1	support
Normal	0.97	0.70	0.82	2213
Exploits	0.82	0.33	0.47	695
Generic	0.35	0.88	0.50	226
Fuzzers	0.31	0.58	0.41	501
Reconnaissance	0.54	0.24	0.33	223
Analysis	0.04	0.69	0.08	16
Backdoor	0.01	0.17	0.02	12
DoS	0.12	0.01	0.02	114

Precision of 1–12% on Analysis/Backdoor/DoS means predictions of these labels are *almost always wrong*; recall of 0.69/0.17 with support $n=16/n=12$ is also **statistically unreliable** at that sample size (a handful of misclassifications swings these numbers enormously) – do not quote these three classes’ metrics without the support column next to them.

- **Training barely converged:** loss went $1.65 \rightarrow 1.43$ over 10 epochs ($\sim 13\%$ reduction) and visibly plateaued after epoch $\sim 3-4$, versus the companion CICIoT run's $1.03 \rightarrow 0.44$ ($\sim 57\%$ reduction). This is consistent with 8-way classification on 8 raw features being a harder problem for this architecture/ epoch budget, not merely "the model needs more of the same training."
- **Class imbalance was only capped, not corrected:** Analysis (333) and Backdoor (257) samples could not reach the `PER_CLASS_CAP=500` used for the other six classes, so the "balanced" training set is still $\sim 1.5-2\times$ smaller for exactly the two worst-performing classes.
- **This run predates the ForwardCache / data-only-Proposition-1 / gradient-once- FGSM fixes.** Proposition 1 here is checked on synthetic Ginibre matrices, not the encoder's real outputs on this dataset – the identity itself is dataset-independent so the number is still valid, but it is not yet consistent with the "real-data-only" methodology adopted for the later CICIoT run.
- **Day 20 (noise-rate sweep, hypothesis H4) was skipped again for this dataset** – H4 (that ε^* is roughly p -independent once fidelities are re-measured at each p) is unconfirmed on UNSW-NB15.
- **H3's "True" is vacuous here too:** $\varepsilon^* = -0.323 < 0$, so any non-negative empirical breakpoint trivially satisfies the comparison – it does not validate Proposition 2 on this dataset.